I. Introduction

In this chapter, we study polynomial approximation of a function in the neighborhood of a point x_0. First, using successive derivatives of a function, we derive Taylor's formulas. We then apply these formulas to obtain finite expansions of a function which simplify the function into a polynomial form near x_0. This approach allows for a precise local study of functions, providing the tools to :

- Calculate indeterminate limits near x_0.
- Determine the tangent to a curve and the curve's position relative to it.
- Identify asymptotic lines and local behavior. (When x_0 is infinite).

Remark : Throughout this chapter, a neighborhood of x_0 (where x_0 may be a finite value or $\pm\infty$) is denoted by $Nr(x_0)$.

II. Taylor's Formulas

II.1 Successive derivatives

Let $f : I \longrightarrow \mathbb{R}$ be a differentiable function, f' be its derivative and x_0 in I.

- If the derivative function $f' : I \to \mathbb{R}$ is itself differentiable at x_0, we say that f is twice differentiable at x_0, and we write
$$\lim_{x\to x_0} \frac{f'(x) - f'(x_0)}{x - x_0} = f''(x_0).$$
- A function $f' : I \to \mathbb{R}$ is twice differentiable if it is twice differentiable at every point in its domain. More generally, we denote :
$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(n+1)} = (f^{(n)})'.$$
- We say that f is of class C^0 on I and write $f \in C^0(I)$ when f is continuous on I.
- We say that f is of class C^n on I with $n \in \mathbb{N}^*$, and write $f \in C^n(I)$ when f is n times differentiable on I, and $f^{(n)}$ is continuous on I.
- We say that f is of class C^∞ on I, and write $f \in C^\infty(I)$, if f is of class C^n on I for every $n \in \mathbb{N}$, and we also say that f is indefinitely differentiable on I.

II.2 Taylor's Lagrange Formula of order n

Theorem 1. *(**Taylor's Lagrange Formula of order** n) Let $f : I \to \mathbb{R}$ be a function and $a, b \in I$ with $a < b$. If f is of class C^n on $[a, b]$ and $f^{(n)}$ is differentiable on $]a, b[$ (the $(n+1)$-th derivative of f exists in $]a, b[$), then there exists $c \in]a, b[$ such that*

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}$$

$$\boxed{f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \frac{f''(a)}{2!}(b-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1} \quad (1)}.$$

- *Formula (1) is called **Taylor's Lagrange formula of order** n.*

- *The sum* $\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k$ *is a polynomial of degree* n *called the Taylor polynomial.*
- *The term* $\frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}$ *is called the Lagrange remainder.*

Example 1. *Let* $f : \mathbb{R} \to \mathbb{R}$ *be the function defined by* $f(x) = \sin(x)$. *According to Taylor's Lagrange Formula of Order 5, there exists* $c \in]0, x[$ *such that*

$$\boxed{\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^6}{6!}\sin(c)}.$$

Indeed, the sine function is of class C^∞ *on* $\mathbb{R}$ *and we have*

$$\begin{aligned}
f(x) = \sin(x) &\implies f(0) = 0 \\
f'(x) = \cos(x) &\implies f'(0) = 1 \\
f''(x) = -\sin(x) &\implies f''(0) = 0 \\
f^{(3)}(x) = -\cos(x) &\implies f^{(3)}(0) = -1 \\
f^{(4)}(x) = \sin(x) &\implies f^{(4)}(0) = 0 \\
f^{(5)}(x) = \cos(x) &\implies f^{(5)}(0) = 1 \\
f^{(6)}(x) = -\sin(x) &\implies f^{(6)}(c) = -\sin(c)
\end{aligned}$$

Substituting into Taylor's Lagrange Formula of order 5, we obtain :

$$\boxed{f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(5)}(0)}{5!}x^5 + \frac{f^{(6)}(c)}{6!}x^6, \quad \text{with } c \in]0, x[}.$$

This yields

$$\boxed{\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^6}{6!}\sin(c) \quad \text{with } c \in]0, x[}.$$

II.3 Taylor's Young formula of order n

Theorem 2. ***(Taylor's Young Formula of order*** n***)***
Let $f : I \longrightarrow \mathbb{R}$ *be a function of class* C^n *on* I *and* $x_0 \in I$, *then there exists a function* $\varepsilon : I \longrightarrow \mathbb{R}$ *satisfying* $\lim_{x \longrightarrow x_0} \varepsilon(x) = 0$ *such that, for all* $x \in I$,

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + (x-x_0)^n \varepsilon(x)$$

$$\boxed{f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + (x-x_0)^n\varepsilon(x) \quad (2)}.$$

- *Formula* (2) *is called* ***Taylor's Young formula of order*** n ***in the neighborhood of*** x_0.
- *The sum* $\sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ *is a polynomial of degree* n *called the Taylor polynomial.*
- *The term* $(x-x_0)^n\varepsilon(x)$ *is called the Young remainder.*

Maclaurin's Formula

If $x_0 = 0$, the Taylor's Young Formula takes the form

$$\boxed{f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + x^n\varepsilon(x) \quad \text{with} \quad \lim_{x \longrightarrow x_0} \varepsilon(x) = 0 \quad (3)}.$$

The formula (3) is called the Maclaurin's formula of order n.

Example 2. *Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a function defined by $f(x) = e^x$. The Young's Taylor Formula of Order n in the neighborhood of 0 (or Maclaurin's formula of order n) gives*

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + x^n\varepsilon(x), \quad \text{with } \lim_{x\to x_0} \varepsilon(x) = 0.}$$

Indeed, the exponential function is of class C^∞ on $\mathbb{R}$, and we have

$$(f(x) = e^x,\ f(0) = 1), \quad (f'(x) = e^x,\ f'(0) = 1), \quad \cdots, (f^{(n)}(x) = e^x,\ f^{(n)}(0) = 1).$$

Substituting into Young's Taylor formula, we get

$$\begin{aligned} f(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + x^n\varepsilon(x) \\ &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + x^n\varepsilon(x). \end{aligned}$$

III. Finite Expansion or Taylor Expansions in the neighborhood of 0

III.1 Definitions and properties

Definition 1.
Let I contain 0. We say that a function f defined on an interval I (possibly except at 0) has a Taylor Expansion or Finite Expansion of order n in the neighborhood of 0 if there exist real numbers $a_0, a_1, \cdots, a_n$ and a function ε defined on I such that :

$$\boxed{f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + x^n\varepsilon(x), \ \text{with } \lim_{x\longrightarrow 0} \varepsilon(x) = 0.}$$

Remark 1.
Note that :

- *The polynomial $P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ has a degree $\leq n$ (denoted $d^\circ P \leq n$) and is called the main or regular part of the limited development.*
- *The expression $x^n\varepsilon(x)$ is also denoted by $o(x^n)$, which is the remainder of the limited development, the negligible part. The notation $o(x^n)$ was introduced by the Russian mathematician Lev Landau.*
  We have $\lim_{x\longrightarrow 0} \frac{o(x^n)}{x^n} = 0$.
- *We write* $f(x) = \underbrace{\sum_{k=0}^{n} a_kx^k}_{\text{main part or regular part}} + \underbrace{o(x^n)}_{\text{remainder}}$.
  - *The remainder $o(x^n)$ is called* ***Little-o notation****.*
  - *By writing $f(x) = P(x) + o(x^n)$, we say that $f(x)$ is equal to the polynomial P plus something that is little-o of x to the n.*
- *In general, we write* **TE** *or sometimes* $\mathbf{TE_n}$ *to indicate a Taylor expansion or finite expansion of order n.*

Example 3.

1. *It is clear that any polynomial function has a TE in $Nr(0)$ of any order. For example :*
   The polynomial function $f(x) = 2+3x+x^3$ has a Taylor expansion in the neighborhood of 0 of order 1, with the main part being $2 + 3x$ and $x^3 = o(x)$ being the negligible part.
   It has a Taylor expansion in the neighborhood of 0 of order 3, with the main part being $2 + 3x + x^3$ and the remainder being zero.

2. *For any $n \in \mathbb{N}$, we have :*

$$1 + x + x^2 + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x} = \frac{1}{1 - x} - \frac{x^{n+1}}{1 - x}.$$

So :

$$\frac{1}{1 - x} = 1 + x + x^2 + x^3 + \cdots + x^n + \frac{x^{n+1}}{1 - x}.$$

Now $\dfrac{x^{n+1}}{1 - x} = x^n \left(\dfrac{x}{1 - x} \right) = o(x^n)$, *as* $\lim\limits_{x \longrightarrow 0} \dfrac{x}{1 - x} = 0$.

Thus, the function $x \longrightarrow \dfrac{1}{1 - x}$ *has a Taylor expansion at* 0 *of order* n *in the form :*

$$\boxed{\frac{1}{1 - x} = 1 + x + x^2 + x^3 + \cdots + x^n + o(x^n)} \tag{1}$$

By writing the function $f(x) = \dfrac{1}{1 + x}$ *in the form* $f(x) = \dfrac{1}{1 - (-x)}$ *and replacing in (1)* (x) *with* $(-x)$, *which is also in the neighborhood of* 0, *we obtain :*

$$\boxed{\frac{1}{1 + x} = 1 - x + x^2 - x^3 \cdots + (-1)^n x^n + o(x^n)}. \tag{2}$$

We have the following result, which follows directly from the definition of a Taylor expansion.

Proposition 1. *If f has a Taylor expansion in the neighborhood of 0, then $\lim\limits_{x \longrightarrow 0} f(x)$ exists and is finite.*

Consequence
If $\lim\limits_{x \longrightarrow 0} f(x)$ does not exist or if it is equal to $\pm\infty$, then f has no Taylor expansion in the neighborhood of 0.

Example 4. *The function $f(x) = \ln x$ has no Taylor expansion in the neighborhood of 0 since $\lim\limits_{x \longrightarrow 0^+} \ln x = -\infty$.*

III.2 Uniqueness of the Taylor expansion

Proposition 2. ***(Uniqueness of the Taylor expansion)***
If f has a Taylor expansion of order n in the neighborhood of 0, then this expansion is unique.

III.3 Taylor expansions and Parity

Proposition 3.
Let f be a function admitting a Taylor expansion of order n in the neighborhood of 0.

- *If f is even, then the coefficients of odd rank in the Taylor expansion are zero.*
- *If f is odd, then the coefficients of even rank in the Taylor expansion are zero.*

III.4 Truncation

Proposition 4.
If a function f has a Taylor expansion of order n in the neighborhood of 0, then it has a Taylor expansion in the neighborhood of 0 of order k, for any $k \leq n$.

Proof :
Suppose that the function f has a Taylor expansion of order n in the neighborhood of 0 in the form :

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k + a_{k+1} x^{k+1} + \cdots + a_n x^n + o(x^n).$$

Then :

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k + x^k \underbrace{\left(x a_{k+1} + \cdots + a_n x^{n-k} + o(x^{n-k})\right)}_{\varepsilon(x)}$$

Since $\lim\limits_{x \longrightarrow 0} \varepsilon(x) = 0$, then :

$$\boxed{f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k + o(x^k)}.$$

Thus, f has a Taylor expansion of order k in the neighborhood of 0.

III.5 Taylor expansions Obtained by the Taylor-Young Formula

Using Formula Maclaurin Formula (3) and setting $(x)^n \varepsilon(x) = o(x^n)$, we obtain the following result :

Proposition 5.
Let $n \in \mathbb{N}$. If a function f is of class C^n in the neighborhood of 0, then it has a Taylor expansion of order n in the neighborhood of 0 given by :

$$\boxed{f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \cdots + \frac{x^n}{n!} f^{(n)}(0) + o(x^n) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k + o(x^n).}$$

In other words : The coefficients of the polynomial in the main part of the Taylor expansion satisfy :

$$\forall n \in \mathbb{N},\ a_n = \frac{f^{(n)}(0)}{n!}.$$

III.5.1 Taylor expansions of Some Elementary Functions Obtained by the Taylor-Young Formula in the neighborhood of 0

1. $f(x) = e^x$, we have $\forall n \in \mathbb{N}$, $f^{(n)}(x) = e^x$ so $f^{(n)}(0) = 1$.

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n)}$$

2. $f(x) = \sin x$, we have $\forall n \in \mathbb{N}$, $f^{(n)}(x) = \sin\left(x + \frac{n\pi}{2}\right)$

so $f^{(n)}(0) = \sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & n = 2k \\ (-1)^k & n = 2k+1 \end{cases}$

$$\boxed{\sin x = x - \frac{x^3}{3!} + \cdots + \frac{(-1)^k}{(2k+1)!} x^{2k+1} + o(x^{2k+2})}.$$

3. $f(x) = \cos x$, we have $\forall n \in \mathbb{N}$, $f^{(n)}(x) = \cos\left(x + \frac{n\pi}{2}\right)$, so

$$\boxed{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{(-1)^k}{(2k)!} x^{2k} + o(x^{2k+1})}.$$

Remark 2.
We notice that in the main part of the Taylor expansion of the function cos, *only even power monomials appear because the function* cos *is even.*
Similarly, for the function sin *which is odd, only odd powers are found in the main part of its Taylor expansion.*

4. Let $f(x) = (1+x)^\alpha$, where $\alpha \in \mathbb{R}$. For any n in $\mathbb{N}$, we have $f^{(n)}(x) = \alpha(\alpha-1)\cdots(\alpha-(n-1))\big(1+x\big)^{\alpha-n}$.
So

$$\boxed{f(x) = (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-(n-1))}{n!} x^n + o(x^n).} \tag{3}$$

III.6 Operations on Taylor expansions :

Let f and g be two functions with Taylor expansions up to order n in the neighborhood of 0, given by :

$$f(x) = P(x) + o(x^n) = P(x) + x^n\varepsilon_1(x), \quad \text{where } \lim_{x\longrightarrow 0} \varepsilon_1(x) = 0 \text{ and } \deg P \leq n.$$

$$g(x) = Q(x) + o(x^n) = Q(x) + x^n\varepsilon_2(x), \quad \text{where } \lim_{x\longrightarrow 0} \varepsilon_2(x) = 0 \text{ and } \deg Q \leq n.$$

Then

1. Sum :

$f + g$ has a Taylor expansion up to order n in the neighborhood of 0 of the form

$$\boxed{f(x) + g(x) = (P(x) + Q(x)) + o(x^n)}.$$

Remark 3.
We have :

$$\boxed{o(x^n) + o(x^n) = o(x^n), \forall n \in \mathbb{N},}$$

because :

$$o(x^n) + o(x^n) = x^n\varepsilon_1(x) + x^n\varepsilon_2(x) = x^n \underbrace{(\varepsilon_1(x) + \varepsilon_2(x))}_{\varepsilon(x)} = x^n\varepsilon(x) = o(x^n) \quad \textit{and} \lim_{x\longrightarrow 0} \varepsilon(x) = 0.$$

Example 5. *(Exercise)*
Determine the Taylor expansion up to order $n = 3$ at $Nr(0)$ of $f(x) = \dfrac{1}{1-x} + e^x$. We have :

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + o(x^3), \; e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3) \; .$$

So

$$f(x) = 2 + 2x + \frac{3}{2}x^2 + \frac{7}{6}x^3 + o(x^3).$$

Example 6.
Determine the Taylor expansion of the function $\cosh x$ and $\sinh x$.

2. Product :

$f \cdot g$ has a Taylor expansion up to order n in the neighborhood of 0 of the form :

$$\boxed{f(x) \cdot g(x) = R(x) + o(x^n)}.$$

where $R(x)$ is the polynomial of degree less than or equal to n obtained by keeping in the product $P(x)$ by $Q(x)$ only the terms of degree less than or equal to n.

(See course supplement for examples).

Example 7.
Calculate the Taylor expansion up to order 4 at $V(0)$ of $\sin x \cdot (1 + \cosh x)$. We have :

$$\begin{aligned} f(x) &= \sin x = x - \frac{1}{6}x^3 + o(x^4) \\ g(x) &= 1 - \cosh x = -\frac{1}{2}x^2 - \frac{1}{24}x^4 + o(x^4) \end{aligned}$$

$$f(x) \cdot g(x) \sin x \cdot (1 - \cosh x) = \left(x - \frac{1}{6}x^3\right) \cdot \left(-\frac{1}{2}x^2 - \frac{1}{24}x^4\right) + o(x^4) = -\frac{1}{2}x^3 + o(x^4).$$

3 Quotient :

If $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + o(x^n)$ and $g(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n + o(x^n)$ and we assume that $\lim\limits_{x\longrightarrow 0} g(x) = b_0 \neq 0$. Then $\dfrac{f}{g}$ has a Taylor expansion in the neighborhood of 0 up to order n, obtained by performing the division according to the increasing powers of x up to order n of P by Q.

Example 8.

1. *Let $f(x) = \dfrac{1+x+x^2}{1+x+x^3}$. Determine the Taylor expansion of the function f in the neighborhood of 0 up to order $n = 3$.*
   The function f is the quotient of the two polynomials $P(x) = 1 + x + x^2 = 1 + x + x^2$ and $Q(x) = 1 + x + x^3$ of degree less than or equal to 3, which coincide with their Taylor expansions up to order 3.
   Performing division up to order 3 according to the increasing powers of x of P by Q :

$$\begin{array}{rrrr|l}
1 & +x & +x^2 & & 1+x+x^3 \\ \cline{5-5}
-1 & -x & -x^3 & & 1+x^2-2x^3 \\ \cline{1-4}
 & x^2 & -x^3 & & \\
 & -x^2 & -x^3 & -x^5 & \\ \cline{2-4}
 & & -2x^3 & -x^5 &
\end{array}$$

   Thus, the Taylor expansion in the neighborhood of 0 up to order 3 is :

$$\frac{1+x+x^2}{1+x+x^3} = 1 + x^2 - 2x^3 + o(x^3).$$

2. *Calculate the Taylor expansion up to order 5 at $V(0)$ of the function $f(x) = \tan x = \dfrac{\sin x}{\cos x}$.*

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5), \quad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^5)$$

   By performing division according to the increasing powers of x, we obtain :

$$\tan x = \frac{\sin x}{\cos x} = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + o(x^5)$$

(See course supplement for remark and examples).

4. Composition :

If f and g have Taylor expansions up to order n in the neighborhood of 0 and if $\lim\limits_{x\longrightarrow 0} g(x) = 0$, then $f \circ g$ has a Taylor expansion up to order n in the neighborhood of 0, obtained by considering in the composite $(P \circ Q)(x)$ only the terms of degree less than or equal to n.

Example 9.
Calculate the Taylor expansion up to order 3 in $Nr(0$ of the function $h(x) = e^{\sin x} = f \circ g$, where $f(x) = e^x$ and $g(x) = \sin x$. Write down first the Taylor expansions up to order 3 of f and g. We have :
$f(x) = e^x = 1 + x + \dfrac{1}{2}x^2 + \dfrac{1}{6}x^3 + o(x^3)$ *and* $g(x) = \sin x = x - \dfrac{1}{6}x^3 + o(x^3)$. *We have* $\lim\limits_{x\longrightarrow 0} g(x) = 0$.
Let $y = \sin x$, $y \longrightarrow 0$ as $x \longrightarrow 0$. Then :

$$\begin{aligned}
e^{\sin x} = e^y &= 1 + y + \frac{1}{2}y^2 + \frac{1}{6}y^3 + o(y^3) \\
&= 1 + \left(x - \frac{1}{6}x^3\right) + \frac{1}{2}\left(x - \frac{1}{6}x^3\right)^2 + \frac{1}{6}\left(x - \frac{1}{6}x^3\right)^3 + o(x^3) \\
&= 1 + \left(x - \frac{1}{6}x^3\right) + \frac{1}{2}x^2 + \frac{1}{6}x + o(x^3)
\end{aligned}$$

We deduce that $h(x) = 1 + x + \frac{1}{2}x^2 + o(x^3)$.

5. Differentiability :

Let f be a differentiable function on an interval I containing 0 with a Taylor expansion up to order n in the neighborhood of 0 of the form $\boxed{f(x) = a_0 + a_1 x + \cdots + a_n x^n + o(x^n)}$. If f' has a Taylor expansion up to order $n-1$ in the neighborhood of 0, then it can be expressed as : $\boxed{f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots + n a_n x^{n-1} + o(x^{n-1})}$.

(See course supplement for examples).

6. Integration :

Proposition 6.

If f has a Taylor expansion of the form $\boxed{f(x) = \sum_{k=0}^{n} a_k x^k + o(x^n)}$ *and F is an antiderivative of f, then F has a Taylor expansion up to order $(n+1)$ in the neighborhood of 0 given by*

$$\boxed{F(x) = F(0) + \sum_{k=0}^{n} a_k \frac{x^{k+1}}{k+1} + o(x^{n+1})}.$$

Example 10.

$f(x) = \frac{1}{1+x}$, $F(x) = \ln(1+x)$, $F(0) = 0$.

$$\boxed{f(x) = \frac{1}{1+x} = 1 - x + x^2 + \cdots + (-1)^n x^n + o(x^n)}.$$

so, by term-by-term integration, we obtain :

$$\boxed{\ln(x+1) = x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{(-1)^n}{n+1} x^{n+1} + o(x^{n+1})}.$$

(See course supplement for other examples).

IV. Taylor expansions around x_0

The Taylor expansion of a function defined in the neighborhood of a point x_0 is obtained by making the variable change $y = x - x_0$ and reducing it to a Taylor expansion around 0. The resulting Taylor expansion takes the following form :

$$\boxed{f(x) = a_0 + a_1(x-x_0)^1 + a_2(x-x_0)^2 + \cdots + a_n(x-x_0)^n + o((x-x_0)^n)}.$$

Example 11.

Let $f(x) = e^x$, calculate the Taylor expansion up to order 3 around $x_0 = 1$. Set $y = x - 1$, so $y \longrightarrow 0$ as $x \longrightarrow 1$. We have :

$$\begin{aligned} f(1+y) &= e^{1+y} = e^1 \cdot e^y \\ &= e^1 \left(1 + y + \frac{y^2}{2} + \frac{y^3}{6} + o(y^3)\right) \end{aligned}$$

Therefore,

$$\boxed{e^x = e^1 + e^1(x-1) + e^1 \frac{(x-1)^2}{2} + e^1 \frac{(x-1)^3}{6} + o((x-1)^3).}$$

(See course supplement for other examples).

V. Taylor expansions at ∞

The Taylor expansion of a function defined in the neighborhood of infinity is obtained by making the variable change $y = \frac{1}{x}$. Note that as x tends to ∞, y tends to 0, thus reducing it to a Taylor expansion around 0. This Taylor expansion is also called an asymptotic expansion.

Example 12.
To calculate the asymptotic expansion around $+\infty$ up to order 2 for $f(x) = \sqrt{\frac{x}{x-1}}$, let's set $y = \frac{1}{x}$, so $y \longrightarrow 0$ as $x \longrightarrow +\infty$. We have :

$$f(\frac{1}{y}) = \sqrt{\frac{\frac{1}{y}}{\frac{1}{y} - 1}} = \sqrt{\frac{1}{1-y}} = (1 + (-y))^{-\frac{1}{2}}.$$

Using the Taylor expansion of the function $(1+x)^\alpha$, for $\alpha = -\frac{1}{2}$, we obtain :

$$\sqrt{\frac{1}{1-y}} = (1 + (-y))^{-\frac{1}{2}} = 1 + \frac{1}{2}y + \frac{3}{8}y^2 + o(y^2).$$

Thus :

$$\boxed{\sqrt{\frac{x}{x-1}} = 1 + \frac{1}{2x} + \frac{3}{8x^2} + o(\frac{1}{x^2}).}$$

VI. Applications of Taylor expansions

VI.1 Equivalent Functions

Definition 2.
Two functions f and g are said to be equivalent in the neighborhood of x_0 (x_0 can be $\pm\infty$) if and only if $\lim_{x \longrightarrow x_0} \frac{f(x)}{g(x)} = 1$. It is denoted as $f \underset{x_0}{\sim} g$.

Example 13.
$\sin x \underset{0}{\sim} x$ *because* $\lim_{x \longrightarrow 0} \frac{\sin x}{x} = 1$.

The Taylor expansion of a function f in the neighborhood of x_0 allows us to find an equivalent to f in the neighborhood of x_0 by taking the first non-zero term of the Taylor expansion around x_0.

Example 14.
The following equivalences are directly deduced from the various calculated Taylor expansions in the previous examples.

1. *We have :* $\ln(1+x) = x - \frac{1}{2}x^2 + o(x^2) \Longrightarrow \ln(1+x) \underset{0}{\sim} x$.
2. *We have :* $\frac{\ln x}{x^2} = (x-1) - \frac{5}{2}(x-1)^2 + \frac{13}{3}(x-1)^3 + o\left((x-1)^3\right)$, *so* $\frac{\ln x}{x^2} \underset{1}{\sim} x - 1$.
3. *We have :* $\sqrt{\frac{x}{x-1}} = 1 + \frac{1}{2x} + \frac{3}{8x^2} + o(\frac{1}{x^2})$, *then* $\sqrt{\frac{x}{x-1}} \underset{+\infty}{\sim} 1$.

VI.2 Limit Calculations

Obtaining a Taylor expansion around x_0 (x_0 can be equal to ∞) of a function f allows for a simple calculation of the limit of this function as x tends to x_0.

Example 15.

$$\lim_{x\longrightarrow 0}\frac{\sin x - x}{x^3} = \lim_{x\longrightarrow 0}\frac{x - \dfrac{x^3}{6} + o(x^3) - x}{x^3} = \lim_{x\longrightarrow 0} -\frac{1}{6} + o(1) = -\frac{1}{6}.$$

(See course supplement for other examples).

VI.3 Tangent Line Equation

Proposition 7.
Suppose f is a function defined at x_0. If f has a Taylor expansion around x_0 (x_0 finite) of the form :

$$f(x) = \underbrace{a_0 + a_1(x - x_0)}_{\textit{Tangent Line Equation}} + \underbrace{a_k(x - x_0)^k}_{\textit{Indicator Term of Position}} + o\left((x - x_0)^k\right), \quad k \in \mathbb{N},\ k \geq 2.$$

Then $y = a_0 + a_1(x - x_0)$ is the equation of the tangent to the curve $(\mathcal{C}_f)$ of f at the point $(x_0, f(x_0))$.

The position of this tangent relative to the curve $\mathcal{C}_f$ depends on the parity of k and the sign of a_k.

- *If k is even and $a_k > 0$, then the curve of f remains locally above its tangent at $(x_0, f(x_0))$.*
- *If k is even and $a_k < 0$, then the curve of f remains locally below its tangent at $(x_0, f(x_0))$.*
- *If k is odd, then the curve of f intersects its tangent at $(x_0, f(x_0))$ (it is an inflection point).*

Example 16.
Determine the equation of the tangent to the curve of the function f defined by $f(x) = e^x$ at the abscissa $x_0 = 1$ and its position relative to the curve $(\mathcal{C}_f)$.
The Taylor expansion of the function f to order 2 around 1 is given by :

$$e^x = e + e(x - 1) + \frac{e}{2}(x - 1)^2 + o((x - 1)^2).$$

We deduce that the line (T) with equation $y = e + e(x - 1)$ is the tangent to the curve $(\mathcal{C}_f)$ at the point $(1, e)$. Since $k = 2$ is even and $a_2 = \dfrac{e}{2} > 0$, the curve $(\mathcal{C}_f)$ is above the tangent (T) in the neighborhood of 1.